

DSC 335, Handout 06: Balance and Sequentiality (*Chapter 4*)

LEARNING OBJECTIVES.

- *Use conjugate models to analyze the effect of different prior distributions/parameters on the posterior distribution.*

Different Priors, Different Posteriors

Example 1. Which of the prior distributions from the racing activity were more informative? Which were more vague?

Example 2. Consider three potential priors on the probability of the hot rod tires being faster than the tiny tires: $\text{Beta}(1,1)$, $\text{Beta}(3,3)$, and $\text{Beta}(10, 1)$.

- (a) Comment on the differences in the prior means and variability.

- (b) Suppose you collect data in which the hot rod tires are faster than the tiny tires 7 out of 10 times. Use `plot_beta_binomial` to create graphs of the relevant distributions, and write a couple of sentences summarizing the effect of the different priors on the posterior.

Different Data, Different Posteriors

Example 3. Suppose three racing teams start with a $\text{Beta}(5,10)$ prior. Team Danica observes 4 out of 10 successes, Team Mario observes 16 out of 40 successes, and Team Lewis observes 40 out of 100 successes. Calculate and compare the posterior distributions, then visualize them in R.

Balance Between Prior and Data

Sequential Analysis

Example 4. Consider a racing scenario where all three teams have a prior of $\text{Beta}(3,7)$. Suppose on Day 1 Team Danica gets 6 out of 10 successes and calculates a posterior distribution. Then on Day 2, they use that posterior as their prior, get 15 out of 20 successes, and calculate a posterior. Team Mario does something similar, but gets 12 out of 15 successes on Day 1 and 9 out of 15 successes on Day 2. Team Lewis gets 21 out of 30 on Day 1 and does not test on Day 2. Will all three teams get the same posterior?

Support and Subjectivity